

THE HODGE CONJECTURE FOR ODD-DEGREE FERMAT FOURFOLDS

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ABSTRACT. Let

$$X_m^4 = \{x_0^m + \cdots + x_5^m = 0\} \subset \mathbf{P}^5$$

be the Fermat fourfold of odd degree m . We prove that the rational cycle class map

$$CH^2(X_m^4)_{\mathbf{Q}} \longrightarrow H^4(X_m^4, \mathbf{Q}) \cap H^{2,2}(X_m^4)$$

is surjective for every odd $m \geq 3$. The proof is uniform in the conductor and reduces every odd conductor to a fixed finite list of endpoint families, which are then realized by explicit cycles. We encode primitive rational Hodge structures by unit orbits of length-six Fermat characters, resolve their exact orders by divisor-lattice Fourier projectors, and prove a descent-or-packet classification. Every terminal packet is realized by an algebraic correspondence built from Fermat-surface divisors, Aoki standard cycles, Ran joins, inverse-pair contractions, and finite power maps. Same-degree surface completion resolves active high-three and high-five orders without imposing divisibility on their complements. A sharp global ℓ^1 ledger then controls ambient one-small completions and the four-face system at the mixed base.

1. INTRODUCTION

The Hodge conjecture for Fermat varieties admits an unusually explicit arithmetic formulation. Primitive cohomology decomposes into one-dimensional eigenspaces for the diagonal automorphism group G_m . A character line $V_m(A)$ has Hodge type $(2, 2)$ precisely when A has age three; the full rational block $V_m[A]_{\mathbf{Q}}$ has this type precisely when every unit translate of A has age three. What is difficult is not detecting Hodge classes but realizing every rational Galois block by algebraic cycles. Decomposable characters are controlled by lower-dimensional Fermat varieties, while standard characters are represented by Aoki's cycles. Composite conductors also contain nonstandard characters for which neither construction applies directly.

Let $e_{[A]} \in \mathbf{Q}[G_m]$ be the rational projector onto $V_m[A]_{\mathbf{Q}}$. If a rational-coefficient algebraic cycle Z has nonzero projection onto $V_m(A)$, then $w = e_{[A]} \text{cl}(Z)$ is a nonzero rational vector in $V_m[A]_{\mathbf{Q}}$. Since this block is a multiplicity-one irreducible rational representation, $\mathbf{Q}[G_m]w = V_m[A]_{\mathbf{Q}}$. The projector and deck translates are algebraic correspondences, so they give algebraic generators of the whole block, as explained in Section 3.

Our result is the following.

Theorem 1.1. *For every odd integer $m \geq 3$, the cycle class map*

$$\text{cl}: CH^2(X_m^4)_{\mathbf{Q}} \longrightarrow H^4(X_m^4, \mathbf{Q}) \cap H^{2,2}(X_m^4)$$

is surjective.

Consequently every odd-degree Fermat fourfold satisfies the rational Hodge conjecture in all codimensions.

The last assertion follows because codimensions zero and four are immediate, codimension one is the Lefschetz $(1, 1)$ theorem, and codimension three follows from hard Lefschetz applied to divisor classes. Thus codimension two is the only substantive case.

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1.1. Relation to earlier work. Shioda [8], Shioda–Katsura [7, Theorem 1.7, pp. 101–102], Ran [6, pp. 138–140], and Aoki [1, Theorem 2-1, p. 388] developed the character calculus and the geometric correspondences used to construct large classes of cycles on Fermat hypersurfaces. Aoki [1, Corollary 2-3, p. 388] proves the Hodge conjecture for Fermat varieties in every dimension whenever $m = p^e$ is a prime power, with no restriction on the exponent. Aoki’s later work [11, Theorem 0.1, p. 177] treats abelian varieties isogenous to factors of powers of the Fermat-curve Jacobian J_m . The Jacobian-power conclusion is recorded in [12, Theorem 1.2, p. 102] for prime powers, twice prime powers, and

$$m = 2^a 3^b 5^c 7^d, \quad c = 0 \text{ or } d = 0,$$

where the exponents are nonnegative. These statements concern Fermat-type abelian varieties and powers of J_m . The relation to hypersurfaces is explicit in [11, Corollary 3.2, p. 184]: algebraicity for the Fermat hypersurfaces in all dimensions is transported to the Fermat-type abelian varieties. The proof of [11, Theorem 0.1, p. 185] first establishes that hypersurface input for the stated degree families. In particular, the odd families $3^a 5^b$ and $3^a 7^b$ are already unbounded precedents.

More recently, da Silva [2, Proposition 3.1, p. 4], in the published version, proved the Hodge conjecture for Fermat fourfolds of degree $m \leq 100$ with $(m, 6) = 1$. It explicitly states that condition P_{33} fails [2, Proposition 3.4, p. 5]. Jumagulov [3, Theorem 1.1, p. 2] proved the result for every odd $3 \leq m \leq 199$, explicitly using an exhaustive machine census [3, Abstract, p. 1]. Its Theorems 1.2–1.4 also give the split, two-pair standard, and level-33-to-66 closure constructions used below. The contribution here is the exhaustiveness of these algebraic leaves for arbitrary odd conductors, established by the short-support, correction, and ambient-completion arguments. For fourfolds this removes the restrictions on prime-factor patterns in the existing unbounded families and covers every odd degree. The present bibliographic and source checks were completed on 2026-09-07.

Schoen [9, Theorem 3.4, pp. 145–146] proves the generalized Hodge conjecture for a selected Hodge substructure on certain generalized Prym varieties under a quasi-simple branching hypothesis. This is an important precedent for transferring algebraic classes from Fermat sources to cyclic-cover geometry, but it does not prove the Hodge conjecture for the Fermat source itself. In the non-quasi-simple case, [9, Remark 3.5, p. 146] instead records the Fermat Hodge conjecture as a sufficient input.

Kang stated a stronger result: [5, Corollary 3.2, p. 227] asserts the generalized Hodge conjecture for Fermat varieties of dimensions three and four in arbitrary degree. Its equation (3.3) invokes the product estimate of [4, Proposition 2.2(iv), pp. 521–522]. We retain Kang’s result as a statement-level precedent, not an input. The proof here is independent of refined motivic dimension and addresses the odd-degree ordinary rational Hodge conjecture directly.

Note on the tensor formula. Write $\mathcal{F}^p H$ for the largest rational sub-Hodge structure of H contained in the Hodge-filtration space $F^p H_{\mathbb{C}}$, and put $\mathcal{F}_t = \mathcal{F}^{-t}$ on homology, as in [4, pp. 520–521]. Here $H_j(X, \mathbb{Q})$ is Borel–Moore homology and W_{-j} is its pure weight- $-j$ part. The product proof [4, p. 522] uses the equality

$$\begin{aligned} & \mathcal{F}_m W_{-i} H_i(X_1 \times X_2, \mathbb{Q}) \\ &= \bigoplus_{j+\ell=i} \mathcal{F}_m (W_{-j} H_j(X_1, \mathbb{Q}) \otimes W_{-\ell} H_{\ell}(X_2, \mathbb{Q})) \\ &= \bigoplus_{j+\ell=i} \bigoplus_{t+s=m} \mathcal{F}_t W_{-j} H_j(X_1, \mathbb{Q}) \otimes \mathcal{F}_s W_{-\ell} H_{\ell}(X_2, \mathbb{Q}). \end{aligned}$$

The second equality fails in general. For a positive-genus curve C , put $H = H^1(C, \mathbb{Q})$. Then $\mathcal{F}^1 H = 0$, whereas the Künneth component of the diagonal is a nonzero rational $(1, 1)$ class in $\mathcal{F}^1(H \otimes H)$. Thus the corresponding factorwise sum $\sum_{a+b=1} \mathcal{F}^a H \otimes \mathcal{F}^b H$ is zero and misses that class; dually the printed equality fails in the $j = \ell = m = 1$ homology component. This identifies a

gap in the product argument, without disproving either the product estimate itself or Kang’s final Hodge-conjecture statement.

1.2. Structure of the proof. Let $A = (a_0, \dots, a_5)$ be a nonzero Fermat character modulo m . Its rational primitive block is the full orbit of A under $U_m = (\mathbf{Z}/m\mathbf{Z})^\times$. The Hodge condition is

$$(1.1) \quad \sum_{i=0}^5 \langle ta_i \rangle_m = 3m \quad (t \in U_m).$$

An inverse pair leaves a Hodge quadruple and is realized by a Fermat-surface divisor, a join, and inverse-pair contraction. We may therefore assume that A is inverse-pair-free.

The proof uses three arithmetic steps.

1. Exact-order Fourier rows and the sharp signed ledger isolate maximal active orders. The vector-valued terminal-kernel classification of Section 2 and the small matrices of Section 6 retain the actual parity and equality cases.
2. Theorem 7.1 resolves every active order divisible by nine, allowing arbitrary five-depth. Theorem 7.3 then resolves every active order divisible by twenty-five by an explicit correction estimate. Corollary 7.4 either gives a same-degree algebraic certificate or removes common content, reducing both small-prime exponents to at most one.
3. At a maximal large part, Theorem 7.5 completes the ambient one-small faces with energies four, five, or six. The mixed base is treated by the four-face system of Section 8 and its zero-top branch in Section 10. The finite endpoints and their geometry are given in Section 9. Every omitted-factor row is used only after the ledger is exhausted or an actual ambient completion has been obtained.

Same-degree surface completion in Lemma 3.3 allows the complementary surface to remain at degree m . Every leaf has an actual Chow certificate, and every change of degree is transported by the finite power-map identity (2). Section 10 assembles the proof without further normal-form or uniqueness assumptions.

1.3. Machine-checked finite certificates. The accompanying Lean 4 sources (Lean 4.32.0, Mathlib) concern finite integer predicates: the cyclic additive matrices of Appendix A, the pure one-small exception levels of Appendix B, and the explicit level-66 identities used in Section 9.2. In particular, `Mixed35Direct.essential_energy` verifies the augmented-energy bound for the enumerated essential representatives; the pure-small certificates identify the complete canonical sets at levels 21, 33, and 39, and the level-66 certificates check the Hodge rows and occurrence identities of the exceptional bridge. The Lean sources, Python comparison scripts, and axiom reports are supplied in the `certificates/` directory. The certificate proofs use the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`, without `sorry` or `native_decide`.

Appendix A is an independent finite certificate. The essential-cross exclusion of Lemma 8.2 is proved analytically through row differences and column Fourier modes, without interpreting the augmented energy as actual mass. The retained finite statements do not formalize the new terminal definition or the infinite dependency chain.

Theorems 1.1, 4.1, and 10.2, the arbitrary-conductor arguments of Sections 6–10, and the Chow realization are ordinary mathematical proofs. Neither the geometric inputs nor the passage from the unit equations to actual cycles is supplied as a Lean theorem. The finite certificates verify their stated finite objects; their success alone does not imply the arbitrary-conductor Hodge theorem.

2. FERMAT CHARACTERS AND RATIONAL HODGE BLOCKS

Let

$$G_m = (\mu_m)^6 / \mu_m^{\text{diag}}$$

act diagonally on X_m^4 . Write a character as

$$A = (a_0, \dots, a_5) \in (\mathbf{Z}/m\mathbf{Z})^6, \quad \sum_i a_i = 0,$$

with each $a_i \neq 0$. The standard Fermat character decomposition and age formula [1, p. 385] give

$$(2.1) \quad H^4(X_m^4, \mathbf{C})_{\text{prim}} = \bigoplus_A V_m(A),$$

where every nonzero summand $V_m(A)$ is one-dimensional. If $\langle a \rangle_m \in \{1, \dots, m-1\}$ denotes the standard representative, then the middle-Hodge condition needed here is

$$(2.2) \quad V_m(A) \subset H^{2,2} \iff \sum_i \langle a_i \rangle_m = 3m.$$

The rational block associated with A is

$$(2.3) \quad V_m[A]_{\mathbf{Q}} \otimes_{\mathbf{Q}} \mathbf{C} = \bigoplus_{t \in U_m / \text{Stab}(A)} V_m(tA).$$

It is a rational Hodge structure of type $(2, 2)$ exactly when (1.1) holds.

More generally, write

$$X_m^d = \{x_0^m + \dots + x_{d+1}^m = 0\} \subset \mathbf{P}^{d+1}.$$

A Hodge tuple of even length L at level m is a multiset of nonzero residues satisfying

$$(1) \quad \sum_{a \in A} \langle ta \rangle_m = Lm/2 \quad (t \in U_m).$$

Its grade is $L/2$. We write $\text{Alg}_m(A)$ for algebraicity of its complete rational primitive block on X_m^{L-2} . All varieties and cycles are over $\mathbf{C}$; rationality refers to cycle coefficients, not to fields of definition of individual subvarieties.

For a subgroup J of a unit group, let E_J denote averaging under its multiplicative action. For $p^e \parallel q$, put

$$J_{p,e} = \begin{cases} U_p, & e = 1, \\ \ker(U_{p^e} \rightarrow U_{p^{e-1}}), & e > 1, \end{cases} \quad P_q = \prod_{p^e \parallel q} (I - E_{J_{p,e}}).$$

We set $P_1 = I$. A terminal fibre is a coset of one of the groups $J_{p,e}$, with every other primary coordinate fixed. The characters surviving P_q are precisely those of conductor q .

We use α_3 for the nonprincipal character of U_3 , normalized by $\alpha_3(1) = 1$ and $\alpha_3(2) = -1$. In a small-coordinate Fourier mode, α_5 denotes a nonprincipal character of U_5 ; the principal small character is written as 1.

Lemma 2.1 (centered tensor spark). *For finite sets X_i of sizes $n_i \geq 2$, put $c_i(x) = \delta_x - n_i^{-1}\mathbf{1}$. The columns $\bigotimes_i c_i(x_i)$ admit no nonzero relation on fewer than $\min_i n_i$ distinct sites. The statement also holds for vector coefficients.*

Proof. The only relation among all centered columns of one factor is their sum. In a proposed relation on fewer than $\min_i n_i$ sites, fewer than n_1 first labels occur, so their columns are independent. Grouping by that label and inducting on the remaining factors proves the assertion. Apply linear functionals to treat vector coefficients. $\square$

Lemma 2.2 (twelve-site classification for large factors). *Let all primes dividing $r > 1$ be at least seven and let V be a characteristic-zero vector space. If $0 \neq h : U_r \rightarrow V$, $P_r h = 0$, and $|\text{supp } h| \leq 12$, then h is constant and nonzero on one complete terminal fibre, or on each of two parallel complete U_7 -fibres. The second possibility has twelve sites and requires $7^1 \parallel r$.*

Proof. Work first on one product of terminal cosets, sorting its sizes $n_1 \leq n_2 \leq \dots$. With at least two factors, $n_1 \geq 6$ and $n_2 \geq 10$. Put $P' = \prod_{i>1} (I - E_i)$. The kernel equation says that $P'h_a$ is independent of the first coordinate a . If one slice is zero, all slices are in $\ker P'$, and each nonzero slice has at least n_2 sites. There is at most one; induction gives a complete fibre. If every slice is nonzero and a singleton slice exists, the difference between it and any other slice has at most $14 - n_1 \leq 8 < n_2$ sites. Lemma 2.1 makes every slice identical to that singleton. If no singleton exists, $2n_1 \leq 12$ forces $n_1 = 6$ and two sites in every slice. Slice differences have at most four sites and vanish, giving two parallel six-fibres. One factor alone has the constant kernel. Finally different terminal-coset blocks are preserved by P_r ; two nonzero blocks under the bound must each be a six-fibre. The only terminal group of size six is U_7 . $\square$

Corollary 2.3 (odd large-factor kernel). *A nonzero integral odd $h \in \ker P_r$ of norm at most twelve is*

$$h = \epsilon(\mathbf{1}_{U_7 \times \{z\}} - \mathbf{1}_{U_7 \times \{-z\}}), \quad r = 7R, \quad R > 1, \quad (R, 7) = 1, \quad \epsilon = \pm 1.$$

In particular its norm is twelve. A nonzero kernel with support at most six is a single U_7 -fibre.

Proof. A fibre is equal to its negative only when it has no other fixed primary coordinate and its varying subgroup contains -1 . This is precisely the full group U_p at a prime level; a constant there is even. Otherwise oddness requires the distinct negative fibre. Apply Lemma 2.2, then integrality and the norm bound. $\square$

Lemma 2.4 (a five-factor under a six-site bound). *On a product with one factor of size five and all other terminal sizes at least six, a nonzero centered-product kernel function on at most six sites is constant on a complete five-fibre or on a complete six-fibre. The same holds for vector coefficients.*

Proof. Group by the five-coordinate. If a slice is zero, every other slice is a residual kernel relation of support at least six, so there is exactly one; its shortest relation is a full six-fibre by the proof of Lemma 2.2. If all five slices are nonzero, one is a singleton and every slice difference has at most three sites. All slices equal that singleton by residual spark. The one-factor case is immediate. $\square$

We will repeatedly use the following explicit detector. For $Q = p^e$, $p \geq 11$, primitive characters of prescribed even parity with exponents 2, 4, or prescribed odd parity with exponents 1, 3, take distinct values at seven. Equality would imply $7^2 = 1 \pmod{Q}$, impossible because $p \nmid 48$. Both choices are primitive, including at higher depth. On a product of such factors, fix all but one character and use these two choices to avoid one specified product value at seven.

Lemma 2.5 (short odd total variation). *Let a permutation T commute with a fixed-point-free involution N , and suppose every T -orbit has length at least three. Let f be integral and satisfy $f(Nx) = -f(x)$, and set*

$$k = \|f\|_1, \quad V(f) = \sum_x |f(Tx) - f(x)|.$$

If $k = 2$, then $V(f) = 4$. If $k = 4$, then $V(f) \geq 4$, and equality means that f consists of two consecutive unit values and their negatives, on one negative-stable cycle or a pair of exchanged cycles. If $k = 6$ and $V(f) = 0$, the support is a pair of exchanged three-cycles with opposite constant unit values.

Proof. On any cycle the variation is at least $2(\max f - \min f)$. On a nonnegative cycle with zeros and maximum $a > 0$ it is at least $2a$. Variations add over cycles. A negative-stable cycle has even length and N acts by its half-period; a nonzero odd function on it has both signs and variation at least four. Exchanged cycles have equal norms and variations.

For $k = 2$, f is a single unit antipodal delta. Its two sites and their translates are distinct, since $Tx = \pm x$ would give an orbit of length at most two. Hence the variation is four. For $k = 4$, twice a single antipodal delta has variation eight. Otherwise the positive and negative parts

each have two unit entries. An exchanged pair has norm two on each cycle; because its length is at least three, variation two is possible only when the two entries have one sign and form a single consecutive plateau. Any second pair contributes at least four more variation. On a negative-stable cycle, variation four permits just one positive and one negative plateau: an additional positive or negative run would add two transitions. The two plateaux are interchanged by N , so their two sites are consecutive. This proves the equality classification. Finally, zero variation means constancy on every cycle. Oddness forces zero on negative-stable cycles. An exchanged pair of nonzero cycles costs $2\ell|a|$ with $\ell \geq 3$, so total norm six forces $\ell = 3$ and $|a| = 1$. $\square$

The characterwise form of Theorem 1.1 is the following.

Proposition 2.6. *If A is a nonzero length-six character modulo an odd integer m and*

$$\sum_i \langle ta_i \rangle_m = 3m \quad (t \in U_m),$$

then the complete rational block $V_m[A]_{\mathbf{Q}}$ is generated by codimension-two algebraic cycles on X_m^4 .

3. THE TYPED CHOW CALCULUS

We use character operations only together with their geometric source, target, codimension, and field of definition. The classical inputs are the Lefschetz (1, 1) theorem, Ran's algebraic join and its primitive injectivity, Aoki's join and inverse-pair contraction, and Aoki's standard-five cycles. The rational block is always the complete unit orbit.

Lemma 3.1 (join, contraction, and finite descent). *Let T_i be middle Fermat characters of length $2d_i$ and grade d_i .*

1. *If the rational blocks of T_1 and T_2 are algebraic, then the block of their ordered juxtaposition $T_1 * T_2$ is algebraic on $X_m^{2(d_1+d_2)-2}$. The induced correspondence adds one Tate factor.*
2. *If $\delta = (u, -u)$ is a nonzero inverse pair and the block of $T * \delta$ is algebraic, then the block of T is algebraic. Repeated inverse pairs may be contracted successively.*
3. *If $n \mid m$, $c = m/n$, and $T = cT_0$ has length $2d$, then algebraicity of the block of T_0 on X_n^{2d-2} is equivalent to algebraicity of the block of T on X_m^{2d-2} .*

Proof. Ran's joining-line incidence correspondence gives the first implication: [6, Proposition 4.6 and Corollary 4.7, p. 140] give injectivity on primitive homology and identify its action on character lines with ordered juxtaposition. The second implication follows directly from Aoki [1, Theorem 1-4(ii), p. 388], applied with the nonnegative even parameter $s = 0$ and the nonzero inverse-pair character δ . Thus the length-two contraction is included in the stated classical theorem. Ran's proof also starts with two zero-dimensional Fermat hypersurfaces.

For the third assertion use

$$\pi_c: X_m^{2d-2} \longrightarrow X_n^{2d-2}, \quad [z_i] \longmapsto [z_i^c].$$

Then

$$\pi_c^* V_n(T_0) = V_m(cT_0), \quad \pi_{c,*} \pi_c^* = c^{2d-1} \text{id}.$$

The map has degree c^{2d-1} . Its quotient group acts trivially on the block of cT_0 , so $\pi_c^* \pi_{c,*}$ is also that nonzero scalar on this block. Hence

$$(2) \quad \text{Alg}_m(cT_0) \iff \text{Alg}_n(T_0).$$

Surjective unit reduction makes both maps compatible with the complete rational orbit. $\square$

The rational group-algebra projector onto a unit orbit is an algebraic rational linear combination of deck correspondences. A nonzero rational vector in this multiplicity-one rational irreducible representation generates it under the deck group. Thus a cycle with nonzero projection on one

complex character line supplies the entire rational block after rational projection and deck translates. No averaging over arithmetic Galois conjugates is required to assert nonvanishing.

Lemma 3.2 (standard packets). *Let $m = 5d$ be odd with $d > 1$, put $p = 5$, and let $S_p(a)$ be the length-six Fermat character*

$$S_p(a) = (a, a + d, \dots, a + (p - 1)d, -pa), \quad a \not\equiv 0 \pmod{d}.$$

The rational block of $S_p(a)$ is algebraic. This includes nonprimitive parameters: with $g = \gcd(a, d)$, the tuple is the common-content inflation by g of the primitive-parameter standard tuple at degree $m/g = 5(d/g)$.

Proof. This is Aoki's standard cycle [1, Theorem 2-1, p. 388]. The standard element range is $d/\gcd(a, d) > 2$ [1, p. 387], and the construction assumes $\gcd(a, d) = 1$. For a nonprimitive parameter with $a \not\equiv 0 \pmod{d}$, divide $g = \gcd(a, d)$. The reduced d/g is odd and greater than one, hence at least three. Apply Aoki's theorem there and return by (2). $\square$

Lemma 3.3 (two same-degree completion rules). *Let A be a Hodge sextuple, and write*

$$L_p(c) = \{c + jm/p : 0 \leq j < p\} \pmod{m}.$$

1. *If $3 \mid m$, $c \not\equiv 0 \pmod{m/3}$, and $A = L_3(c) \uplus R$, then $\text{Alg}_m(A)$ holds.*
2. *If $5 \mid m$, $c \not\equiv 0 \pmod{m/5}$, $d \in L_5(c)$, and $A = (L_5(c) \setminus \{d\}) \uplus \{y, z\}$, then $\text{Alg}_m(A)$ holds. No divisibility of y, z is assumed.*

The second rule gives a uniform fibre criterion for the two-pair standard closure of Jumagulov [3, Theorem 1.3, p. 4]: the two pairs and the standard sextuple plus surface quadruple are exactly those in (5). The criterion places the complementary surface at the ambient degree. The new exhaustiveness arguments below determine when these established geometric closures must occur at an arbitrary odd conductor.

Proof. The elementary fibre identity is

$$(3) \quad \sum_{x \in L_p(c)} \langle tx \rangle_m = \frac{p-1}{2}m + \langle ptc \rangle_m.$$

It follows by reducing to the p equally spaced lifts of tc modulo m/p . For the first assertion put

$$S = L_3(c) \uplus \{-3c\}, \quad B = R \uplus \{3c\}.$$

Both are nonzero Hodge quadruples by (3) and (1). The literal identity

$$(4) \quad A \uplus (3c, -3c) = S \uplus B$$

proves the claim by joining their surface divisor classes and contracting the displayed pair. The join first reaches codimension three on X_m^6 ; the contraction returns to codimension two on X_m^4 . For the second assertion set

$$S = L_5(c) \uplus \{-5c\}, \quad B = (5c, y, z, -d).$$

Here S is standard-five, while (1) and (3) give grade two for B . All entries are nonzero, and

$$(5) \quad A \uplus (d, -d) \uplus (5c, -5c) = S \uplus B.$$

Join the standard cycle and the divisor to codimension four on X_m^8 , then contract twice to codimension two on X_m^4 . If c is not primitive, divide $\gcd(c, m/5)$ before applying Aoki and use (2) to return. The hypothesis $c \not\equiv 0 \pmod{m/5}$ makes the reduced standard block nonzero. $\square$

Remark 3.4 (same-degree terminal certificates). The surface B in either rule is at the same level m . The terminal condition in Section 4 therefore allows these actual surface/standard join-and-contraction certificates at degree m , without requiring the surface factor to be inflated from $m/3$ or $m/5$. No divisibility condition on the complementary entries is part of either completion rule.

4. THE CONDUCTOR-UNIFORM CLASSIFICATION

Call a Hodge sextuple *decomposable* if it contains an inverse pair. A nondecomposable sextuple is called *terminal* when it has one of the following actual Chow certificates: joins of same-degree Fermat-surface divisor blocks followed by inverse-pair contractions; an Aoki standard-five block, optionally joined with a same-degree surface divisor and contracted; a split block obtained from a divisor on a product of Fermat curves with complementary character types; or the exceptional level-33 construction of Section 9.2. The definition includes unit and permutation translates and finite power-map transports of these certificates. In particular, both completion rules of Lemma 3.3 are terminal certificates at degree m ; no prior inflation of their surface factor from $m/3$ or $m/5$ is required.

Theorem 4.1 (descent or packet). *Let $m \geq 3$ be odd and let A be a nonzero length-six character satisfying the unit equations (1.1). Then one of the following holds.*

1. A is decomposable;
2. A is terminal;
3. there exist a proper divisor $n \mid m$, $c = m/n > 1$, and a Hodge sextuple A_0 modulo n such that $A = cA_0$.

Every terminal case in the second alternative has an actual Chow realization. In particular, the alternatives are stable under passage to the complete rational unit orbit.

The proof of Theorem 4.1 is given in Section 10 after the high-depth, ambient one-small, and mixed-base arguments. Proposition 5.2 records the well-founded degree changes. Assuming the classification for the moment, the main theorem follows.

Proof of Theorem 1.1 from Theorem 4.1. Apply Theorem 4.1 to a rational Hodge block. A decomposable character is algebraic by Lemma 3.1 and the Lefschetz $(1, 1)$ theorem on the Fermat-surface factor. A terminal character is algebraic by Lemmas 3.1–3.3 and the explicit packet constructions. In the remaining case, the finite power-map identity (2) reduces the claim to a proper divisor of the conductor. Strong induction on m proves Proposition 2.6 and hence the surjectivity of the primitive cycle class map. The nonprimitive part of H^4 is generated by the square of the hyperplane class. $\square$

5. EXHAUSTIVE BRANCH STRUCTURE AND TERMINATION

The conductor bookkeeping separates the arithmetic arguments while retaining the full ambient Hodge sextuple. Maximality isolates rows whose conductor contains the selected large part; actual completion or a sharp ledger equality controls the other supports.

Proposition 5.1 (branch coverage). *Every odd conductor $m \geq 3$ belongs to exactly one of the following regimes.*

1. If $15 \nmid m$, either all prime divisors are at least seven, or a unique $s \in \{3, 5\}$ divides m . In the latter case

$$(5.1) \quad m = s^e M, \quad (M, s) = 1, \quad p \geq 7 \text{ for every } p \mid M.$$

The possibilities $M = 1, e = 1$, $M = 1, e \geq 2$, $M > 1, e = 1$, and $M > 1, e \geq 2$ are disjoint and exhaustive.

2. If $15 \mid m$, then uniquely

$$(5.2) \quad m = 3^a 5^b M, \quad a, b \geq 1, \quad (M, 15) = 1.$$

The cases $(a, b) = (1, 1)$, $a \geq 2, b = 1$, $a = 1, b \geq 2$, and $a, b \geq 2$ are disjoint and exhaustive.

Proof. This is unique factorization, with three and five distinguished. Theorem 7.1 applies to active three-depth at least two in any of these regimes, and Theorem 7.3 subsequently handles active five-depth at least two. The partition introduces no compatibility assumption between two primary walls. $\square$

Proposition 5.2 (well-founded descent). *Every nonterminal change of degree used in Theorem 4.1 removes a common content $c > 1$ and therefore strictly lowers the positive conductor. In particular, for $m = 3^a 5^b M$, Corollary 7.4 reduces to*

$$(5.3) \quad 3^{\min(a,1)} 5^{\min(b,1)} M$$

unless a same-degree leaf has already been obtained. At a reduced level, if no nontrivial large part is active, all entries descend to degree three, five, or fifteen. Any such passage from a larger degree is again strict. Consequently no reduction cycle exists.

Proof. If $A = cA_0$ and $c > 1$, then $m/c < m$. Surjective unit reduction preserves the Hodge equations, and (2) transports algebraicity in both directions. The positive integers are well ordered. $\square$

Proposition 5.3 (maximal rows and incomparable supports). *At a level with three- and five-adic exponents at most one, choose a divisibility-maximal active large part r . A Fourier row with large conductor r receives contributions only from large parts divisible by r . Maximality removes proper multiples, but does not itself remove incomparable supports.*

The local classification gives a contradiction, an actual terminal certificate, or a grade-two core whose two ambient complements form an inverse pair. In the one-small sector, energy four gives that core, energy five gives the forced level-33 completion, and energy six exhausts the sextuple. Thus globalization follows from actual completion or ledger exhaustion.

Proof. Equation (7) proves the restriction on contributing orders. Theorem 7.5 proves the ambient one-small completions. Lemmas 8.1–8.4 resolve the nonzero mixed face, and Section 10 treats the two-small and empty sectors. Each use of an omitted-factor row there follows an explicit twelve-unit equality; the lower-energy cases instead complete the actual Hodge tuple before excluding external supports. $\square$

6. EXACT-ORDER FOURIER ROWS AND SMALL MATRICES

An inverse pair in a Hodge sextuple leaves a Hodge quadruple, so that case is algebraic by Section 3. We henceforth assume inverse-pair-freeness. For an exact order $q \mid m$, define

$$\mu_q(x) = \#\{a \in A : a = (m/q)x\}, \quad F_q(x) = \mu_q(x) - \mu_q(-x), \quad x \in U_q.$$

At odd m the sharp ledger is

$$(6) \quad F_q(-x) = -F_q(x), \quad \sum_{q \mid m, q > 1} \|F_q\|_1 = 12.$$

Signed vanishing implies actual emptiness, because an inverse pair is forbidden. Use the nonconjugate Fourier convention

$$\widehat{f}(\chi) = \sum_x f(x)\chi(x), \quad B_q(x) = \langle x \rangle_q / q - \frac{1}{2}, \quad C_q(\chi) = \sum_{x \in U_q} B_q(x)\chi(x).$$

Then $T_s f(y) = f(sy)$ has multiplier $\overline{\chi(s)}$; in particular its support is the inverse image under multiplication by s . If χ is a primitive odd character of conductor f , the standard Dirichlet identities give

$$C_f(\chi) = -L(0, \chi) \neq 0.$$

Indeed $L(0, \chi) = -i\tau(\chi)L(1, \bar{\chi})/\pi$, with $L(1, \bar{\chi}) \neq 0$ and $|\tau(\chi)|^2 = f$; see [10, equations 25.15.5, 25.15.9, 25.15.10, and 27.10.11]. The exact-order equation is

$$(7) \quad \sum_{f|q|m} \frac{C_q(\bar{\chi})}{\varphi(q)} \hat{F}_q(\chi) = 0.$$

To derive it, rewrite (1) as $\sum_{q,x} F_q(x)B_q(tx) = 0$, and take its normalized Fourier coefficient against $\bar{\chi}(t)$. An order- q term vanishes unless $f \mid q$; otherwise the substitution $u = tx$ gives $\chi(x)C_q(\bar{\chi})/\varphi(q)$. Unit reduction to any divisor is surjective. For $p \nmid f$, induction of a nonprincipal character satisfies

$$(8) \quad C_{pf}(\chi) = (1 - \chi(p))C_f(\chi).$$

This follows by subtracting the multiples of p from $\sum_{a=1}^{pf} a\chi(a)$ and dividing by pf ; the sum of χ over a period is zero. Thus at a maximal active exact order, (7) and oddness give $P_q F_q = 0$. A row only sees orders divisible by its conductor. Incomparable supports are not discarded until a ledger equality proves their emptiness or an actual ambient completion resolves the sextuple.

Lemma 6.1 (small matrix classification). *An integral additive 3×4 matrix of norm at most six consists of complete three-columns, a complete four-row, an essential mass-five matrix, or an essential mass-six matrix. For 3×5 , the alternatives are complete three-columns, a complete five-row, or an essential mass-six matrix. In the latter cases the signed coefficient sums are respectively $\pm 1, \pm 2$ for 3×4 , and ± 2 for 3×5 . An integral additive 2×5 matrix of norm at most six is row-invariant of even mass, or has mass five with exactly one unit entry per column.*

Proof. Write entries as $u_i + v_j$, choosing integer potentials. In the three-row case the norm of each column is at least $\max u_i - \min u_i$, so with four or five columns that integral range is at most one. Range zero gives complete three-columns. Otherwise, up to a gauge, row permutation, and overall sign, use $u = (1, 0, 0)$. A column is $(t+1, t, t)$. The norm bound permits only $t = 0, -1$, since any other column has cost at least four, making the total at least seven with four columns. With n columns and b columns of the second kind, the mass is $n + b$. Thus $b \leq 2$ for $n = 4$ and $b \leq 1$ for $n = 5$. The coefficient sums follow at once: $3 - 2 = 1$, $2 - 4 = -2$, and $4 - 2 = 2$. A row-invariant matrix of mass six may contain a three-column with multiplicity two, which is not omitted. For two rows, the row difference d has $5|d| \leq 6$. If $d = 0$, the mass is even. If $d = \pm 1$, each column has odd norm, so the total is odd and at least five; under the bound it equals five, with one unit entry in every column. $\square$

Lemma 6.2 (residual boundary with row-lift freedom). *Let a signed multiplicity restricted to an oriented terminal block have norm at most six, and suppose its small centered projector followed by the large residual projector vanishes. Either every residual slice satisfies the small kernel equation, or all six sites form a residual U_7 -fibre and the projected small coefficient vector is constant on that fibre. If all small factor sizes are at least three, all small labels and signed coefficients are constant. For small sizes 2, 5, the five-label is fixed and the quantity $\sigma\alpha_3(a)$ is constant; the two-row label a itself need not be fixed.*

Proof. If there is no residual factor, its projector is the identity and the small equation holds immediately. Otherwise apply the residual projector to the vector-valued small projection. Fewer than six residual sites give zero by Lemma 2.1; exactly six give a constant-vector complete U_7 -fibre. Equality in the original integral norm leaves one unit coefficient at each site. Centered delta columns for a set of size at least three are pairwise nonproportional. Equality of nonzero simple tensors consequently fixes all labels and coefficients in that case. On a two-set the two centered columns are negatives; multiplying by its nontrivial character records precisely that sign freedom. This gives the stated 2, 5 alternative rather than an unjustified fixed-row assertion. $\square$

7. HIGH PRIMARY DEPTH AND AMBIENT ONE-SMALL COMPLETION

7.1. Direct high-three closure.

Theorem 7.1 (any active three-depth at least two is algebraic). *Let m be odd and A an inverse-pair-free Hodge sextuple. If some entry has exact additive order divisible by nine, then A contains a complete three-fibre, a standard-five sextuple, or four members of a nonzero five-fibre. In particular $\text{Alg}_m(A)$ follows from Lemma 3.3 and Aoki's standard cycle.*

Proof. Choose a divisibility-maximal active order q divisible by nine. Every active proper multiple would again be divisible by nine, so its primitive row is isolated. Write $q = 3^i 5^j r$, where $i \geq 2$ and $(r, 15) = 1$; j may be zero. The terminal three-kernel has size three and odd order, so a terminal block and its negative are disjoint. Each oriented block has integral norm at most six.

Apply Lemma 6.2. A residual boundary exhausts all six physical entries, fixes their three-primary label up to a common sign, and forces all entries to have exact order q . Divide the common content m/q . Modulo 3^i their sum is $\pm 6z$, with z a unit. This is nonzero because $i \geq 2$. Thus no such boundary occurs.

All residual slices therefore satisfy the small equation. If $j = 0$, a nonzero slice is constant on a three-fibre. If $j = 1$, use the 3×4 case of Lemma 6.1; if $j \geq 2$, use 3×5 . A complete signed three-column gives an actual complete three-fibre (choose its positive or negative orientation), and is algebraic by Lemma 3.3. A five-row at depth $j \geq 2$ gives five actual members of $L_5(c)$; the sixth is $-5c$ by the zero-sum congruence. A four-row when $j = 1$ gives four actual members of $L_5(c)$ and is covered by the same-degree completion rule, regardless of its two complements.

An essential mass-six slice exhausts the ledger. All its three-primary labels are congruent to one unit modulo three, but its signed coefficient sum is ± 2 . The six actual entries, after dividing common content, therefore have nonzero sum modulo three, a contradiction.

Only the essential mass-five 3×4 slice remains. Its five effective labels have sum S that is a unit modulo every prime dividing q . At three the coefficient sum is ± 1 . At five, if b_0 is the exceptional column, the weighted sum is $\pm 3b_0$. At each residual prime the common label is multiplied by ± 1 . Hence the sixth actual entry is $-(m/q)S$ and also has order q . Reapply the just-established six-mass classification to this full stratum. If the sixth entry lay in a new independent small slice, that slice would contain only one unit entry, contrary to the minimum-support bound for the centered-product kernel in Lemma 2.1. A three-fibre or incomplete five-fibre is an algebraic leaf; the essential six-mass or residual boundary is impossible. Thus every alternative gives an asserted leaf or contradiction. $\square$

Corollary 7.2. *Unless A is already algebraic by one of these explicit certificates, every active order has three-adic valuation at most one. Hence, if $3^a \parallel m$, all coordinates are divisible by $3^{\max(a-1, 0)}$.*

This conclusion allows arbitrarily large ambient five-depth. It eliminates every active high-three order before the high-five argument is applied.

7.2. High-five closure and the correction estimate.

Theorem 7.3 (high-five closure after high-three removal). *Let m be odd and let A be an inverse-pair-free Hodge sextuple with no active order divisible by nine. If some active order is divisible by twenty-five, then $\text{Alg}_m(A)$ has a standard-five or surface-join-contraction certificate.*

Proof. Choose a maximal active base $h = q/3^{v_3(q)}$ among orders divisible by twenty-five. Write $h = 5^j r$, $j \geq 2$, $(r, 15) = 1$. Only the orders h and $3h$ can contribute to a conductor- h row. Set $D = F_h$ and $F = F_{3h}$, defining the latter to be zero if $3h \nmid m$. If $F \neq 0$, its primitive row gives the $U_3 \times J_5$ small equation with the residual projector.

First eliminate the residual boundary in Lemma 6.2. It uses all six entries. Let η_3 be the odd character of U_3 , choose a primitive even character θ at five-depth j , and primitive even characters on

every other remaining primary factor. Twisting θ by the quadratic character modulo five preserves both evenness and full depth, and changes its value at seven by -1 . Thus one of the two resulting odd characters of conductor $3h/7$ has a nonzero omitted-seven Euler factor. Its coefficient of F is nonzero: on the positive fibre the product $\sigma\eta_3(a)$ is fixed by Lemma 6.2, and the negative fibre gives the same contribution. The isolated row (7) contradicts this.

Now Lemma 6.1 applies on every small slice. A complete five-row is a standard-five leaf. An essential mass-five slice has five actual effective labels whose sum is a unit at every primary factor: its three-coordinate weighted sum is $\pm 5u$, and its signed coefficient sum on the five and residual factors is ± 1 or ± 3 . The sixth entry therefore has the same exact order $3h$. A six-mass additive 2×5 slice is row-invariant; another noninvariant slice would already cost five, and a separate nonzero slice costs at least two. Thus, after the standard leaves and contradictions, the entire top face is

$$F(a, y) = v(y), \quad v(-y) = -v(y), \quad k = \|v\|_1 \in \{2, 4, 6\}.$$

Its full signed norm is $2k$. The edge row gives

$$(9) \quad PC = 0, \quad C = D - (T_3v - v), \quad P = (I - E_{J_5}) \prod_{p^e \| r} (I - E_{J_{p,e}}).$$

We now prove $C = 0$ by a norm estimate.

Put $g = T_3v - v$. The ledger gives $\|D\|_1 \leq 12 - 2k$ and $\|g\|_1 \leq 2k$, so $\|C\|_1 \leq 12$. All terminal blocks are exchanged in pairs under negation because J_5 has odd order. On an oriented half, C has norm at most six. Lemma 2.4, integrality, and the budget show that a nonzero correction must have the exact form

$$(10) \quad C = \epsilon(\mathbf{1}_L - \mathbf{1}_{-L}), \quad \epsilon = \pm 1, \quad |L| = \rho \in \{5, 6\}.$$

There is just one such antipodal block pair. A six-fibre is in an exponent-one U_7 -direction. In either case L has a fixed nonzero five-primary residue z modulo 5^{j-1} .

Let $\Lambda = L \cup (-L)$. A source pair $\{x, -x\}$ of v contributes to g at $x, -x, x/3, -x/3$. At most one of its two antipodal pairs can lie in Λ , since otherwise $3z = \pm z \pmod{5^{j-1}}$. Consequently

$$(11) \quad \|g|_\Lambda\|_1 \leq k, \quad \|D\|_1 = \|C + g\|_1 \geq 2\rho - k.$$

For $\rho = 6$, this contradicts $\|D\|_1 \leq 12 - 2k$ for every $k > 0$. For $\rho = 5$ it contradicts that bound when $k \geq 4$. When $\rho = 5, k = 2$, v is a single unit antipodal delta. If g meets Λ , its other antipodal pair lies outside Λ and contributes two unavoidable units. Hence $\|D\|_1 \geq 10 - 2 + 2 = 10 > 8$; if there is no meeting the bound is even larger. This excludes (10) and proves $C = 0$.

Thus

$$(12) \quad D = T_3v - v, \quad k + \frac{1}{2}\|T_3v - v\|_1 \leq 6.$$

Every multiplication-by-three orbit in U_h has length at least twenty, since $5^j \mid h$: the order at five is four, and $v_5(3^4 - 1) = 1$ gives $\text{ord}_{5^j}(3) = 4 \cdot 5^{j-1}$ by binomial lifting. These are point orbits in U_h . For $k = 2$, $v = \pm(\delta_c - \delta_{-c})$ and the variation is four. The two top atoms and two forced lower atoms form a Hodge quadruple; the other two entries have grade one and are opposite. For $k = 4$, Lemma 2.5 gives variation at least four; equality is forced. On exchanged orbits, norm two and variation two on one orbit give two adjacent unit values; on a negative-stable orbit the same follows from its positive and negative plateaux. Therefore, up to translation and sign,

$$v = \delta_c + \delta_{3c} - \delta_{-c} - \delta_{-3c}.$$

For precision about the actual labels, put $d = m/(3h)$ and let P_c be the two unit lifts of $c \in U_h$ to level $3h$, multiplied by d . Then

$$S_c = P_c \uplus \{3d(c/3), -3dc\} = L_3(dc) \uplus \{-3dc\}$$

is a Hodge quadruple. The two S -blocks satisfy

$$S_c \uplus S_{3c} = A \uplus (3dc, -3dc).$$

This realizes the six-entry equality. For $k = 6$ the variation is zero; an odd invariant function vanishes on negative-stable orbits, and a nonzero pair of exchanged orbits would have norm at least forty. Hence this case is impossible.

It remains to justify the case $F = 0$. Then $PD = 0$. An oriented nonzero block has at most six units, so Lemma 2.4 gives a complete five-fibre (standard-five) or six U_7 -points with fixed five-primary unit label. The latter exhausts all entries and has sum $\pm 6z \not\equiv 0 \pmod{5^j}$ after removal of common content. Thus it is impossible, completing the proof. $\square$

Corollary 7.4 (simultaneous primary-depth reduction). *For odd $m = 3^a 5^b M$, $(M, 15) = 1$, a Hodge sextuple is algebraic by an explicit same-level leaf certificate, or all its entries are divisible by*

$$c = 3^{\max(a-1,0)} 5^{\max(b-1,0)}.$$

In the latter case $A = cA_0$ is a Hodge sextuple at level

$$m/c = 3^{\min(a,1)} 5^{\min(b,1)} M.$$

Proof. Remove inverse pairs first. Theorem 7.1 eliminates any active nine-divisible order unless a leaf is obtained. Theorem 7.3 then does the same for twenty-five. The remaining valuation bounds are exactly the common-content assertion. Dividing preserves all unit equations by surjective unit reduction; use (2) for algebraicity. $\square$

7.3. Ambient one-small completion. All primes of $r > 1$ are at least seven and $s \in \{3, 5\}$. Suppose at a maximal active large part only the exact-order faces sr and r are present. This applies both to the pure one-small setting and to the mixed base with its other two faces zero. Write

$$F(a, y) = F_{sr}(a, y), \quad D(y) = F_r(y), \quad e = \frac{1}{2}(\|F\|_1 + \|D\|_1), \quad F \neq 0.$$

The top and edge rows are isolated by maximality; rows omitting a large factor are not yet isolated. Formula (7) proves their independence of the ambient modulus. In unnormalized form the common rescaling from level sr to m is $[m/(sr)]\varphi(m)/\varphi(sr)$.

Theorem 7.5 (ambient completion). *Under these hypotheses and the six-entry Hodge ledger, one obtains an actual energy-four Hodge quadruple whose complements are inverse, the energy-five prefix of a level-33 exceptional packet whose last entry is forced, or an energy-six algebraic packet exhausting all entries.*

Proof. First suppose r is not a prime. Apply $\pi_s = I - E_{U_s}$ to the small-coordinate vector and set $H(y) = \pi_s F(\cdot, y)$. Then $P_r H = 0$ and $H(-y) = -N_s H(y)$, where N_s negates the small coordinate. A nonzero H under twelve units, by Lemma 2.2, must use two opposite U_7 -fibres and exhaust $\|F\|_1 = 12$. A self-opposite single fibre would require r prime. For $s = 3$ the projected coefficient $\sigma\alpha_3(a)$ is constant along a fibre; for $s = 5$ the map $(\sigma, a) \mapsto \pi_5(\sigma\delta_a)$ is injective, so its small label and sign are constant. Choose an odd nontrivial α_s detecting this coefficient and a primitive even residual character on $r/7$ avoiding the one omitted-seven value. The detector following Lemma 2.4 provides it. After the ledger exhaustion the conductor- $sr/7$ row has one nonzero term and nonzero Euler factor. Contradiction. Thus

$$F(a, y) = b(y), \quad b \text{ integral and odd.}$$

The edge row gives $P_r C = 0$, where $C = D - (T_s - 1)b$. If $s = 5$, $k = \|b\|_1 \geq 2$ and

$$\|C\|_1 \leq 12 - (s - 3)k \leq 8,$$

so Corollary 2.3 forces $C = 0$. If $s = 3$, the only possible nonzero correction has norm twelve and is a pair of opposite U_7 -fibres. Equality in

$$\|C\|_1 \leq \|D\|_1 + \|T_3b - b\|_1 \leq 12 - 2k + 2k = 12$$

forces $D, T_3b - b$ to have disjoint support inside the unit-coefficient support of C , and forces b, T_3b to have disjoint support there too. Both supports would have residual labels $z, -z$ on $r/7$, requiring $3z = \pm z$. All primes of $r/7$ are at least eleven, so this is impossible. Hence also $C = 0$.

The energy is now

$$(13) \quad e = \frac{s-1}{2}k + \frac{1}{2}\|T_sb - b\|_1 \leq 6.$$

For $s = 5$, necessarily $k = 2$, the variation is four, and the complete packet is standard-five. For $s = 3, k = 2$ gives a four-entry three-standard surface block. For $k = 4$, Lemma 2.5 makes variation four equivalent to two consecutive unit values and their negatives, so $e = 6$ and the two-surface identity from the high-five proof applies. For $k = 6$, invariance would require $\text{ord}_r(3) \leq 3$ on a nonzero odd orbit. Orders one or two are impossible; order three forces $r \mid 26$, hence $r = 13$, contrary to the current nonprime case. This completes the nonprime argument.

Now let $r = p$ be prime. The exact top kernel gives, in the zero-mean gauge,

$$(14) \quad F(a, y) = u(y) + b(a), \quad u(-y) = -u(y), \quad b(-a) = -b(a).$$

Integrality makes the potentials either both integral or both strict half-integral: all differences within each potential are integral and oddness forces twice the common fractional part to be integral. The edge equation gives

$$D = T_s u - u$$

exactly, since the only P_p -kernel is constant and this correction is odd.

For $s = 5$, write $b = (c_1, c_2, -c_1, -c_2)$. Pairing opposite small coordinates gives

$$\sum_a |u(y) + b(a)| = 2 \max(|u(y)|, |c_1|) + 2 \max(|u(y)|, |c_2|).$$

If the potentials are integral, nonzero noninvariant u has norm at least two and variation at least four, giving $e \geq 6$. Nonzero invariant odd u costs more: $\text{ord}_p(5) \geq 3$. When $u = 0$, nonzero b already gives $e \geq p - 1 \geq 6$. In the half-integral case the same lower bound $e \geq p - 1$ holds. Thus $e = 6$ and every other stratum is empty. The remaining small-only row consequently says $(T_p - 1)b = 0$. An integral nonzero b would force $p = 7$, equality, $u = 0$, and support on a single antipodal small pair; multiplication by $7 \equiv 2 \pmod{5}$ does not preserve it. In the half-integral case equality is possible only at $p = 7$, but would require an odd nowhere-zero u invariant under five, although $5^3 = -1 \pmod{7}$. Hence $b = 0$. Then u is an integral single antipodal delta and the packet is standard-five.

For $s = 3$ put $b = (c, -c)$ and $V(u) = \|T_3u - u\|_1$. The exact energy is

$$(15) \quad e = \sum_{y \in U_p} \max(|u(y)|, |c|) + \frac{1}{2}V(u) \leq 6.$$

In the integral case, $c \neq 0$ forces $p = 7$, $|c| = 1$, and equality. Then $V(u) = 0$ and $3^3 = -1 \pmod{7}$ give $u = 0$. If $c = 0$, the even norm $k = \|u\|_1$ is 2, 4, or 6. Norm two gives an actual four-entry surface block; norm four has variation at least four, and equality gives the quasi packet. Norm six requires three-invariance, hence $p = 13$ as above, with energy six.

In the strict half-integral case $p \leq 13$, so $p \in \{7, 11, 13\}$. Also $|c| = 1/2$. At $p = 11, 13$, a value of $|u| \geq 3/2$ already increases the first term by at least two and exceeds the budget. At $p = 7$, such a value would give first term at least five and variation at least six, also impossible. Thus every $|u(y)| = 1/2$. At $p = 13$ the first term is already six. At $p = 11$ it is five, and a nonconstant sign pattern on either five-cycle has total variation at least four on the negative pair of cycles; thus u is three-invariant and $e = 5$. At $p = 7$, U_7 is a three-generated cycle with negation its half-period.

Its three half-period signs have one or three sign changes on the path ending at the negative of the first; the full variations are two or six and the energies are four or six. This proves that only $e = 4, 5, 6$ occur.

We finally justify actual ambient completions. At energy four with $c = 0$ the actual core is the explicit three-standard quartet. At energy four in the half-integral $p = 7$ case, the core has zero full 21- and 7-conductor rows by (14) and $D = T_3u - u$. Its remaining odd row has conductor three and vanishes because $1 - \alpha_3(7) = 0$. Fourier inversion therefore proves that these four actual entries form a Hodge quadruple. Inflating to the ambient level preserves this assertion. Subtracting its Hodge equations from the sextuple leaves a grade-one pair, necessarily opposite.

At energy five, $p = 11$. The five positive entries form one small row times a coset of $\langle 3 \rangle \subset U_{11}$. Their sum is 11 or 22 modulo 33. Thus the ambient zero-sum equation forces the sixth entry to have order three, and the completed tuple is a unit/permutation translate of

$$(m/33)(1, 4, 16, 22, 25, 31).$$

At energy six, the two faces exhaust all six entries. All actual entries are multiples of $m/(3p)$, so the full effective Hodge equations hold at $3p$. For $p > 13$ the preceding integral classification already gave the quasi family. The remaining exact levels 21, 33, 39 are the finite lists in Section 9, and their blocks are algebraic by the constructions there. No omitted-large-factor row has been isolated before one of these actual completions. $\square$

8. THE EXACT FOUR-FACE SYSTEM AT LEVEL $15M$

Assume $m = 15M$, $(M, 15) = 1$. At a maximal active large part $r \mid M$ write its four faces as

$$F_{35}(a, b, y), \quad F_3(a, y), \quad F_5(b, y), \quad D(y), \quad a \in U_3, \quad b \in U_5, \quad y \in U_r.$$

Here the subscripts abbreviate the exact orders $15r, 3r, 5r, r$. We write $E_s = E_{U_s}$, and N_s for negation of the U_s -coordinate. For a primitive large character β , the top equation is

$$(16) \quad P_r(I - E_3)(I - E_5)F_{35} = 0.$$

The edge equations, obtained from (7), are

$$(17) \quad 4\hat{F}_3(\alpha_3, \beta) + (1 + \overline{\beta(5)})\hat{F}_{35}(\alpha_3, 1, \beta) = 0,$$

$$(18) \quad 2\hat{F}_5(\alpha_5, \beta) + (1 - \overline{\alpha_5(3)\beta(3)})\hat{F}_{35}(1, \alpha_5, \beta) = 0.$$

The trivial-small equation is

$$(19) \quad \begin{aligned} 0 = & 8\hat{D}(\beta) + 4(1 - \overline{\beta(3)})\hat{F}_3(1, \beta) + 2(1 - \overline{\beta(5)})\hat{F}_5(1, \beta) \\ & + (1 - \overline{\beta(3)})(1 - \overline{\beta(5)})\hat{F}_{35}(1, 1, \beta). \end{aligned}$$

Parity ensures the even modes cause no additional conditions.

Lemma 8.1 (Pointwise additive reduction). $F_{35}(a, b, y) = u(a, y) + v(b, y)$ with integral potentials.

Proof. Set $D'_b(y) = F_{35}(1, b, y) - F_{35}(2, b, y)$ and $H(y) = \pi_5 D'(y)$, where $\pi_5 = I - E_5$. Then $P_r H = 0$, $D'(-y) = N_5 D'(y)$, and $H(-y) = N_5 H(y)$. Thus $\text{supp } H$ is negative-stable, but no parity is imposed on an individual rectangular difference. If $r = 1$, $H = 0$ immediately.

Suppose $H \neq 0$. Since $|\text{supp } H| \leq \|F_{35}\|_1 \leq 12$, Lemma 2.2 applies. A single negative-stable fibre requires $r = p$ and H constant. Then $N_5 H = H$, so every integral representative $D'(y)$ is constant on antipodal small pairs and has norm at least two. Consequently $\|F_{35}\|_1 \geq 2(p - 1)$; equality forces $p = 7$, norm twelve, and

$$D'(y) \in \{\epsilon \mathbf{1}_{\{1,4\}}, -\epsilon \mathbf{1}_{\{2,3\}}\}, \quad D'(-y) = D'(y).$$

If two fibres occur, they are opposite U_7 -fibres in $r = 7R$, $R > 1$, and already require all twelve units of F_{35} . There is one unit entry over each residual site. Injectivity of $(\epsilon, b) \mapsto \pi_5(\epsilon\delta_b)$ then gives

$$D'(u, z) = \epsilon\delta_b, \quad D'(u, -z) = \epsilon\delta_{-b} \quad (u \in U_7).$$

These descriptions retain the possible row choice of each original F_{35} atom; the row difference, not a fictitious fixed row, is used below.

Either alternative exhausts (6), so all other actual strata vanish. The omitted-seven row is now isolated. In the lone $r = 7$ case take $\chi = \alpha_3\lambda_5$, where λ_5 is quadratic. The row difference gives $\hat{F}_{35}(\chi) = 12\epsilon \neq 0$, while $\chi(7) = -1$. In the paired case choose a primitive odd β on R and the two odd quartic characters $\gamma_{\pm}$ of U_5 . Then $\chi_{\pm} = \alpha_3\gamma_{\pm}\beta$ are odd, both detect the full multiplicity with coefficient $12\epsilon\gamma_{\pm}(b)\beta(z)$, and have values $\pm i\beta(7)$ at seven. At least one value is not one. In either case (8) and primitive odd nonvanishing contradict (7). Thus $H = 0$. Fixing one row and one column supplies integral additive potentials directly. $\square$

Lemma 8.2 (essential-cross exclusion). *For $r > 1$, no additive slice of F_{35} genuinely varies in both the row and column directions.*

Proof. At an oriented residual pair $\{y, -y\}$ the small matrix has physical mass at most six. Its integral common row difference d satisfies $\|F_y\|_1 \geq 4|d|$ and the norm is even. Thus an essential slice has $d = \pm 1$ and mass four or six. A mass-two slice is row-invariant.

Define $G_3(y) = F_3(1, y) - F_3(2, y)$. Equation (17) gives

$$P_r(G_3 - h_3) = 0, \quad h_3 = -(I + T_5)d.$$

The essential slice and its negative give the same unit value of d at $y, -y$, so h_3 is a nonzero constant c on the four distinct sites $\{\pm y, \pm y/5\}$. If the total top mass is six, $F_3 = 0$; the only possible additional mass-two slice does not change d . A four-site kernel relation contradicts Lemma 2.1.

Otherwise the top mass is four and at most two physical atoms remain. The correction $K = G_3 - h_3$ is even and has support at most eight. Lemma 2.2 and parity give $K = 0$ or $r = 7$ with K constant. If $K = 0$, the four-site G_3 consumes both remaining atoms. Choose a nonprincipal column character α_5 detecting the nonconstant column potential w at y . Its forced edge mode is

$$h_5 = \overline{\alpha_5(3)}T_3w - w,$$

nonzero on four distinct residual sites. With no F_5 atom remaining, this contradicts the scalar spark.

If $r = 7$ and $K = \lambda 1 \neq 0$, support at most four for G_3 forces $\lambda = -c$. Then G_3 is supported on the complementary antipodal pair and consumes at least one physical atom, leaving at most one for F_5 . A single-atom F_5 mode has at most two sites. It cannot differ from h_5 by a constant on U_7 : a nonzero constant would require the four existing values of h_5 all to be equal, since its two absent sites would already consume the two allowed sites. For the quadratic character h_5 is odd and not constant; for a quartic character its values at y and $y/3$ have ratio $-\overline{\alpha_5(3)} = \pm i$. A zero constant leaves four sites. Contradiction. $\square$

Appendix A records the independent finite classification of these additive matrices and its augmented-energy inequality. Lemma 8.2 uses the row-difference and column Fourier modes above; it does not interpret augmented energy as actual physical mass.

If any row-only slice occurs, it has exactly four atoms. An additional mass-two column-only slice would make the top mass six and contradict the same G_3 argument. Hence the four atoms form a complete U_5 -row, that is, four members of $L_5(c)$. Lemma 3.3 resolves it directly. The case $r = 1$ descends to the finite level fifteen in Section 9. We are left with the global column-only sector

$$(20) \quad F_{35}(a, b, y) = v(b, y), \quad k = \|v\|_1 > 0, \quad \|F_{35}\|_1 = 2k.$$

8.1. Correction lifting and small-constant closure. Put

$$(S_3v)(b, y) = v(3b, 3y), \quad g = S_3v - v, \quad C = F_5 - g.$$

Equation (18) gives $P_r(I - E_5)C = 0$, and

$$(21) \quad \|C\|_1 \leq \|F_5\|_1 + 2k \leq 12 - \|F_3\|_1 - \|D\|_1.$$

Lemma 8.3 (correction is small-coordinate constant). *In (20), under the full Hodge system (16)–(19) and ledger (6), $C(b, y) = c(y)$, with integral odd c .*

Proof. For $r = 1$ this is immediate. Let $H(y) = \pi_5 C(\cdot, y)$; now its actual parity is $H(-y) = -N_5 H(y)$. If $H \neq 0$, Lemma 2.2 applies using $\|C\|_1 \leq 12$. In a paired-fibre case, equality leaves one unit coefficient of C per residual site. The injective projected delta map fixes one small column on the first U_7 -fibre and its negative column on the opposite fibre. Thus C uses just one antipodal small pair. A single self-opposite fibre requires $r = p$. Then $C(b, y) = a(b) + f(y)$ in the odd zero-mean gauge. The potentials are both integral or both half-integral, and

$$\|C\|_1 \geq (p - 1)\|a\|_1 \geq 2(p - 1)$$

when $a \neq 0$. Hence $p = 7$ and every bound is an equality. In the integral case a is a unit antipodal delta and $f = 0$, again using only two small columns. In the half-integral case every potential value has absolute value $1/2$, and

$$C = \epsilon(\mathbf{1}_{R \times Z} - \mathbf{1}_{-R \times (-Z)}),$$

where $R \subset U_5$ and $Z \subset U_7$ are transversals of the antipodal pairs, of sizes two and three.

Every case has $\|C\|_1 = 12$. Equality in (21) makes $F_3 = D = 0$, $\|g\|_1 = 2k$, and every nonzero coefficient of C a unit. Equality in the triangle inequality then forces F_5 and g to be disjoint, and v, S_3v to be disjoint with both supports inside $\text{supp } C$. A two-column support cannot contain both, since multiplication by three interchanges the two antipodal column pairs. In the half-integral rectangle, $\text{Agg}_5 C = 2\epsilon(\mathbf{1}_Z - \mathbf{1}_{-Z})$ is nonzero and odd, where Agg_5 sums the four columns. Choose an odd character β on U_7 detecting it. Since five generates U_7 , $\beta(5) \neq 1$. The equality already exhausted all other strata. Substituting $F_5 = S_3v - v + C$ and $F_{35} = v$ in (19) cancels the v -terms and leaves

$$2(1 - \overline{\beta(5)})\widehat{\text{Agg}_5 C}(\beta) = 0,$$

a contradiction. Thus $H = 0$ and the assertion follows. $\square$

The top plus five-face mass satisfies $2k + \|F_5\|_1 \geq 8$. For $k \geq 4$ this is immediate. For $k = 2$, v is a unit antipodal delta, and g has one unit in each of four distinct residual slices. Adding a column-constant integer cannot reduce any such slice below norm one. At most four signed units remain for F_3, D ; its row difference is a P_r -kernel function with support at most four. Hence

$$F_3(a, y) = b_3(y)$$

by Lemma 2.1. The trivial-small row first gives only

$$P_r K = 0, \quad K = D - (T_3 - 1)b_3 - (T_5 - 1)c.$$

Writing $J = 2k + \|F_5\|_1$, we have

$$4\|c\|_1 \leq J, \quad J + 2\|b_3\|_1 + \|D\|_1 \leq 12,$$

so $\|K\|_1 \leq 12 - J + 2\|c\|_1 \leq 8$. For $c \neq 0$ use integral oddness $\|c\|_1 \geq 2$ and $J \geq 4\|c\|_1$; for $c = 0$ use $J \geq 4$. Corollary 2.3 therefore gives $K = 0$.

Lemma 8.4 (small-constant closure). *$c = b_3 = D = 0$ in the column-only sector.*

Proof. Assume $c \neq 0$. Then $4\|c\|_1 \leq 12$ and integral oddness give $\|c\|_1 = 2$. Since $J \geq 8$, the remaining budget allows either $b_3 = 0$, in which case $D = (T_5 - 1)c$ has norm four, or $\|b_3\|_1 = 2, D = 0$. In both cases $J \leq 8$. Thus

$$8 = \|C\|_1 \leq \|F_5\|_1 + \|g\|_1 \leq \|F_5\|_1 + 2k = J \leq 8.$$

As above, integrality and unit coefficients force v, S_3v to be disjoint and supported in C . Its residual support is just $\{z, -z\}$, so $3z = \pm z$ would follow, impossible. Thus $c = 0$. Now $D = (T_3 - 1)b_3$. If $b_3 \neq 0$, the cost $2\|b_3\|_1 + \|(T_3 - 1)b_3\|_1$ is at least eight: norm two has variation four and norm at least four suffices already. The top and five-face cost is also at least eight, violating the ledger. Hence $b_3 = D = 0$. $\square$

Finally $k + \frac{1}{2}\|S_3v - v\|_1 \leq 6$. Every S_3 orbit has length divisible by four, owing to its U_5 coordinate. Norm two gives an actual surface core and an inverse complementary pair. Lemma 2.5 turns norm four and variation four into two adjacent unit values and their negatives, and the same two-surface contraction identity used earlier. Norm six would require invariance; a negative-stable orbit then contributes zero and an exchanged pair would cost at least eight. Thus the nonzero mixed face always has an algebraic certificate.

9. FINITE ENDPOINTS AND THEIR CHOW REALIZATIONS

9.1. Complete finite orbit lists. At level fifteen the inverse-pair-free unit/permutation orbits have representatives

$$(1, 2, 7, 11, 12, 12), \quad (1, 4, 7, 10, 10, 13).$$

The second is standard-five. With $C(w) = L_3(w) \uplus \{-3w\}$, the first obeys

$$(1, 2, 7, 11, 12, 12) \uplus (6, 9) = C(2) \uplus C(6).$$

For $p = 7, 11, 13$, put

$$Q_p = (1, 1 + p, 1 + 2p, 3 + p, 3 + 2p, 3p - 9) \pmod{3p}.$$

The complete inverse-pair-free orbit lists are

p	representatives
7	$Q_7, (1, 4, 9, 15, 16, 18), (1, 4, 10, 13, 16, 19),$
11	$Q_{11}, (1, 4, 16, 22, 25, 31),$
13	$Q_{13}, (1, 7, 16, 22, 34, 37), (1, 14, 16, 22, 29, 35).$

An exhaustive enumeration sets the sixth entry to $3m$ minus the sum of the first five nondecreasing positive entries below m , retains its legal nondecreasing range, excludes inverse pairs, and tests every unit equation. Sorting every unit translate gives canonical orbits. Comparison of the complete canonical sets gives the displayed lists. The unquotiented counts at 15, 21, 33, 39 are 6, 12, 14, 24; their orbit counts are 2, 3, 2, 3. Appendix B retains the exact finite certificate at 21, 33, 39.

9.2. Chow realization of the finite leaves. The Q_p blocks satisfy

$$Q_p \uplus (3, -3) = C(1) \uplus C(3)$$

and are algebraic. The split closure is the one in Jumagulov [3, Theorem 1.2, p. 4]. The four split representatives admit

$$\begin{array}{l|l} 21 & (1, 4, 16) * (9, 15, 18), \quad (1, 4, 16) * (10, 13, 19), \\ 39 & (1, 16, 22) * (7, 34, 37), \quad (1, 16, 22) * (14, 29, 35). \end{array}$$

Each triple sums to zero at its level. Since every unit translate of the juxtaposition is Hodge, the two curve characters have complementary types; their diagonal rational orbit is a $(1, 1)$ -block on the product of curves. Lefschetz supplies a divisor there and Ran's algebraic join [6, Section 4, pp. 138–139] maps it nontrivially to the fourfold block [6, Proposition 4.6 and Corollary 4.7, p. 140].

This uses the divisor on the product; algebraicity of either curve character alone is not asserted. The rational projector argument in Section 3 gives the complete orbit.

For the exceptional level-33 tuple $w = (1, 4, 16, 22, 25, 31)$, we use the certificate of Jumagulov [3, Theorem 1.4, p. 5]. The following $w, a, Q, S, \beta, \gamma, \delta_0, \delta_1$ are the same tuples as there, and the level-66 completion, contractions, and descent by the squaring map are the same construction. Work at level 66 with $a = 2w$ and

$$\begin{aligned} a &= (2, 8, 32, 44, 50, 62), & Q &= (1, 25, 44, 62), \\ S &= (2, 8, 32, 41, 50, 65), & \delta_0 &= (1, 65, 25, 41), & \delta_1 &= (33, 33). \\ \beta &= (2, 32, 33, 65), & \gamma &= (8, 33, 41, 50), \end{aligned}$$

The three quadruples Q, β, γ are Hodge and

$$a \uplus \delta_0 = Q \uplus S, \quad S \uplus \delta_1 = \beta \uplus \gamma.$$

Join β, γ to codimension three on X_{66}^6 and contract δ_1 to obtain the codimension-two block of S . Then join with Q to codimension four on X_{66}^8 and contract the two pairs in δ_0 , returning to codimension two on X_{66}^4 . The self-pair $(33, 33)$ is a legitimate nonzero inverse pair at even level 66. This uses only that explicit even auxiliary variety. The squaring map $X_{66}^4 \rightarrow X_{33}^4$ has quotient group of order 32 acting trivially on a ; hence $\pi^*\pi_* = 32$ on that block and its algebraic image at level 33 is nonzero. Reduction $U_{66} \simeq U_{33}$ preserves the full orbit. The finite certificate checks all displayed Hodge conditions and multiset identities exactly.

10. THE ZERO MIXED FACE AND GLOBAL ASSEMBLY

Retain the notation of Section 8 and first suppose $r > 1$ and $F_{35} = 0$. If exactly one of F_3, F_5 is nonzero, Theorem 7.5 applies. If both are zero, $P_r D = 0$; an odd nonzero large-factor kernel would have norm twelve by Corollary 2.3. This exhausts all strata and its omitted-seven row is detected by a primitive odd character on $r/7$ avoiding one value at seven. Thus it is impossible.

Suppose both small faces are nonzero. For r not prime, put $H_s = \pi_s F_s$. Its negative-stable support is strictly smaller than twelve, since the other nonzero odd face already costs at least two units. Lemma 2.2 gives $H_s = 0$: a single self-opposite fibre requires r prime, and a paired fibre requires twelve sites. For $r = p$ prime, write $F_s(a, y) = c_s(a) + b_s(y)$ with odd integral or half-integral potentials. A nonzero c_5 costs at least $(p-1)\|c_5\|_1 \geq 12$ signed units, impossible with $F_3 \neq 0$. Thus $c_5 = 0$ and $F_5 = b_5$ costs at least eight. A nonzero c_3 then costs at least $p-1 \geq 6$ signed units, again impossible. Consequently in every case

$$F_3 = b_3(y), \quad F_5 = b_5(y).$$

Their signed costs are at least four and eight. Equality in the ledger makes each b_s a single antipodal delta and empties D and every other stratum. Only now are all residual rows isolated. Put

$$K = (T_3 - I)b_3 + (T_5 - I)b_5.$$

The primitive odd rows of exact conductor r , whose Bernoulli coefficients are nonzero, give $P_r K = 0$. The function K is integral and odd, and

$$\|K\|_1 \leq 2\|b_3\|_1 + 2\|b_5\|_1 = 8.$$

By Corollary 2.3, a nonzero integral odd function in $\ker P_r$ has norm at least twelve. Hence $K = 0$, proving

$$(22) \quad (T_3 - 1)b_3 + (T_5 - 1)b_5 = 0.$$

On the basis $e_y = \delta_y - \delta_{-y}$, indexed by $U_r/\{\pm 1\}$, each summand is an edge between distinct vertices. Equality of the two edges forces one of

$$r \mid 5 - 3, \quad r \mid 5 + 3, \quad r \mid 15 - 1, \quad r \mid 15 + 1.$$

Every prime of r is at least seven, so $r = 7$ is the only nontrivial possibility. The equations then normalize to $b_3 = e_3, b_5 = e_1$. The actual level-105 representative is

$$(3, 24, 50, 66, 85, 87) = (L_5(3) \setminus \{45\}) \uplus \{50, 85\},$$

which is the example in Jumagulov [3, Theorem 1.3, p. 4] and is algebraic by Lemma 3.3. Unit translations and inflation cover the whole branch.

Proposition 10.1 (no small prime). *If every prime dividing an odd m is at least seven, every Hodge sextuple is a union of inverse pairs.*

Proof. For an inverse-pair-free input, take a maximal active exact order q . Corollary 2.3 makes the only nonzero possibility a pair of opposite U_7 -fibres at $q = 7R$, exhausting the ledger. Choose a primitive odd character on R with value at seven different from one, using the detector in Section 2. Its coefficient is nonzero and (8) contradicts the single surviving row. After removing inverse pairs from a general sextuple, any nonempty remainder has even length at most four. Its odd ledger is at most eight, immediately excluded by the same kernel argument. Hence no remainder exists. $\square$

Theorem 10.2 (assembled odd-degree conclusion). *The classical geometric inputs in Section 3 and the arithmetic arguments above give*

$$\text{cl} : CH^2(X_m^4)_{\mathbf{Q}} \rightarrow H^4(X_m^4, \mathbf{Q}) \cap H^{2,2}(X_m^4) \quad \text{for every odd } m \geq 3.$$

Proof. By the classical character decomposition it suffices to realize each Hodge sextuple. An inverse pair is a surface-divisor leaf. Otherwise apply Corollary 7.4. Its explicit leaves are algebraic, or its finite power descent gives a Hodge tuple at a level with three- and five-adic exponents at most one. If neither small prime occurs, Proposition 10.1 applies. If exactly one s occurs, choose a maximal active large part r . For $r > 1$, a nonzero F_{sr} is covered by Theorem 7.5; if it is zero the sole F_r is excluded by the odd-kernel and omitted-seven argument. For $r = 1$, all entries descend to the prime level s , where $P_s F_s = 0$ and oddness force $F_s = 0$, contradicting activity. If both small primes occur, use the four-face arguments in Sections 8 and 10. At $r > 1$ they give an algebraic leaf or contradiction. At $r = 1$ every entry has order dividing fifteen; remove common content and apply the finite level-fifteen classification in Section 9. Every leaf uses actual cycles by Section 3 or Section 9; every passage back to the original level uses (2). The nonprimitive part is the square of the hyperplane class. $\square$

Proof of Theorem 4.1. The preceding proof gives the stated classification as well as algebraicity. An inverse pair is the first alternative. Corollary 7.4 either supplies a same-degree terminal certificate or removes common content; when that content exceeds one, this is the third alternative. At the reduced level, the maximal-large-part arguments of Theorem 7.5 and Sections 8 and 10 give an actual terminal certificate or a contradiction. If the maximal large part is one, removing common content reduces to degree three, five, or fifteen. The first two are impossible for an inverse-pair-free input, and degree fifteen has the two explicit certificates of Section 9. All certificates are preserved by unit translation, permutation, and finite power-map transport. This proves Theorem 4.1 with the terminal definition of Section 4. $\square$

APPENDIX A. THE 2×4 CYCLIC ADDITIVE CERTIFICATE

We retain the complete finite cyclic additive certificate. Its augmented energy is a finite matrix invariant; the analytic essential-cross exclusion in Lemma 8.2 does not identify it with actual mass. Order the four U_5 -columns by the multiplication-by-three cycle

$$(1, 3, 4, 2).$$

Let $M = (m_{ij}) \in \mathbf{Z}^{2 \times 4}$ satisfy

$$(A.1) \quad m_{ij} = u_i + v_j, \quad 0 < \|M\|_1 \leq 6.$$

Equivalently, after flattening by rows, the three equations

$$(A.2) \quad m_{1j} - m_{10} = m_{0j} - m_{00}, \quad j = 1, 2, 3,$$

hold. The exact equivalence group is generated by the row swap, cyclic rotation of the four columns, and multiplication of all coefficients by -1 . Reflections and arbitrary column permutations are not allowed, because they do not preserve the multiplication-by-three dynamics.

There are two exhaustive enumerations. The first visits the full integer ℓ^1 -ball in eight coordinates:

$$\#\{x \in \mathbf{Z}^8 : \|x\|_1 \leq 6\} = 40081$$

and retains the 288 nonzero solutions of (A.2). The second writes

$$(A.3) \quad \begin{aligned} (m_{00}, m_{01}, m_{02}, m_{03}) &= (a, a + c_1, a + c_2, a + c_3), \\ (m_{10}, m_{11}, m_{12}, m_{13}) &= (a + d, a + d + c_1, a + d + c_2, a + d + c_3). \end{aligned}$$

The norm bound implies

$$|a| \leq 6, \quad |d|, |c_j| \leq 12,$$

so enumeration in these ranges is independently exhaustive. The two canonical class sets agree. They contain twenty-nine classes, with norm profile

$$(A.4) \quad \#\{\|M\|_1 = 2, 4, 6\} = (1, 9, 19).$$

A class is essential when both $d \neq 0$ and the four entries of its first row are not all equal. For a vector w , set

$$(A.5) \quad q(w) = \min_{c \in \mathbf{Z}} \sum_i |w_i + c|.$$

Then

$$(A.6) \quad q_{\text{row}} = |d|, \quad q_{\text{col}} = q(m_{00}, m_{01}, m_{02}, m_{03}).$$

The eleven essential representatives and their augmented energies

$$(A.7) \quad E(M) = \|M\|_1 + 2q_{\text{row}} + 2q_{\text{col}}$$

are as follows. Each displayed vector is flattened by rows.

M	$\ M\ _1$	$(q_{\text{row}}, q_{\text{col}})$	$E(M)$
$(-1, -1, -1, 0; 0, 0, 0, 1)$	4	(1, 1)	8
$(-1, -1, 0, 0; 0, 0, 1, 1)$	4	(1, 2)	10
$(-1, 0, -1, 0; 0, 1, 0, 1)$	4	(1, 2)	10
$(-2, -1, -1, -1; -1, 0, 0, 0)$	6	(1, 1)	10
$(-2, -1, -1, 0; -1, 0, 0, 1)$	6	(1, 2)	12
$(-2, -1, 0, -1; -1, 0, 1, 0)$	6	(1, 2)	12
$(-2, -1, 0, 0; -1, 0, 1, 1)$	6	(1, 3)	14
$(-2, 0, -1, -1; -1, 1, 0, 0)$	6	(1, 2)	12
$(-2, 0, -1, 0; -1, 1, 0, 1)$	6	(1, 3)	14
$(-2, 0, 0, -1; -1, 1, 1, 0)$	6	(1, 3)	14
$(-2, 0, 0, 0; -1, 1, 1, 1)$	6	(1, 2)	12

Thus every essential class has $E(M) \geq 8 > 6$. This is the independent finite inequality certified by the table. As a diagnostic, quotienting incorrectly by the full symmetric group on four columns leaves only sixteen classes and collapses thirteen genuine cyclic classes.

The supplementary verifier `verify_mixed35_c4_additive_classification.py` uses only integer arithmetic. It asserts equality of the two enumerations, recomputes (A.4)–(A.7), and fixes the canonical payload by SHA-256

6884de68c535501c5d08250dbb0d297fda791e05c767291166bce208600127dd.

The hash is an integrity check on the finite table, not evidence for any of the analytic arbitrary-conductor steps in Sections 6–10.

APPENDIX B. THE PURE ONE-SMALL EXCEPTION LEVELS

We give the finite classification used after the analytic one-small energy and ambient-transport arguments have reduced the pure tail to

$$(B.1) \quad m = 3p, \quad p \in \{7, 11, 13\}.$$

The equivalence group is exactly

$$(B.2) \quad U_{3p} \times \mathfrak{S}_6.$$

The symmetric group is removed by sorting the sextuple; multiplication by -1 is already contained in U_{3p} and is not divided out again.

Put

$$(B.3) \quad Q_p = (1, 1 + p, 1 + 2p, 3 + p, 3 + 2p, 3p - 9).$$

This represents the quasi orbit. Up to (B.2), the complete lists of inverse-pair-free grade-three Hodge sextuples are

$$(B.4) \quad \begin{array}{c|l} p & \text{orbit representatives} \\ \hline 7 & Q_7, (1, 4, 9, 15, 16, 18), (1, 4, 10, 13, 16, 19), \\ 11 & Q_{11}, (1, 4, 16, 22, 25, 31), \\ 13 & Q_{13}, (1, 7, 16, 22, 34, 37), (1, 14, 16, 22, 29, 35). \end{array}$$

The second orbit at $p = 11$ is the level-33 exceptional block, whose algebraicity certificate is due to Jumagulov [3, Theorem 1.4, p. 5] (arXiv:2608.18134). The remaining nonquasi representatives are precisely the split packets appearing in Section 9.2.

Here is an exhaustive certificate for (B.4). For fixed $m = 3p$, sort the six entries and enumerate the first five by combinations with repetition. The $t = 1$ Hodge row determines the last entry uniquely:

$$(B.5) \quad a_6 = 3m - \sum_{i=1}^5 a_i.$$

Retain it only when $a_5 \leq a_6 < m$, discard every tuple containing an opposite pair, and test

$$(B.6) \quad \sum_{i=1}^6 \langle ta_i \rangle_m = 3m \quad (t \in U_m).$$

Finally replace every survivor by the lexicographically least sorted unit translate. This visits every unordered sextuple once before quotienting and therefore proves exhaustiveness. The numbers retained before taking unit orbits, followed by the orbit counts, are

$$(B.7) \quad \begin{array}{c|cc} p & \text{retained sextuples} & \text{unit orbits} \\ \hline 7 & 12 & 3 \\ 11 & 14 & 2 \\ 13 & 24 & 3. \end{array}$$

Direct comparison of the canonical sets gives exactly (B.4).

The supplementary verifier

```
calculations/bundles/fermat-level15m-pure-small-exception-levels/
verify_pure_small_exception_levels.py
```

uses only integer arithmetic and asserts the canonical-set equality, not merely the counts (B.7). On OMEN it returns

`pure_exception_levels=PASS.`

In the preceding energy reduction, energy four contains an actual grade-two block; the two remaining entries have grade one and are forced to be opposite, so this case returns to the decomposable branch. Energy five occurs only at $p = 11$: the strict-half-integral prefix is unique and the ambient sum determines the sixth entry, giving the level-33 exceptional representative in (B.4). The energy-six survivors are the quasi and split packets listed there. Their Chow realizations are those already used in Section 9.2.

Appendix B certifies only the effective levels (B.1). The arbitrary- M transport, full-depth residual-factor detectors, and four-face globalization are the analytic arguments in the body of the paper.

APPENDIX C. BRANCH, BUDGET, AND GEOMETRIC DEPENDENCIES

Table 1 records the dependencies of Theorem 10.2, separately from the finite certificate lists in Section 9 and Appendices A–B. Write $\delta_u = (u, -u)$, and let B denote an actual Hodge quadruple. In the last column, a sequence such as $2, 2 \rightarrow 6 \rightarrow 4$ records Fermat dimensions at the stated level: two surface-divisor sources, their join, and inverse-pair contraction back to the fourfold. The direct standard-five cycle has codimension two on X_m^4 ; joining it with a surface divisor and contracting twice gives $4, 2 \rightarrow 8 \rightarrow 4$. Ran’s primitive injectivity and Aoki’s Theorem 1-4(ii) give nonvanishing for these operations (Lemma 3.1), and Aoki’s Theorem 2-1 gives it for the standard cycle (Lemma 3.2).

The condition $r = 1$ concerns the maximal *active* large part; it does not assert that the ambient large part M is one. In every geometric row, the full rational block is obtained by applying the rational deck projector of Section 3 to a cycle with nonzero character-line projection and taking its deck translates.

Table 1: Branches, budgets, and geometric certificates.

Branch	Maximal active order	Twelve-unit budget	When omitted-factor rows are available	Actual multiset identity or outcome	Source–target; nonvanishing
Inverse pair	None needed.	Resolved before the inverse-pair-free ledger (6).	Not used.	$A = B \uplus \delta_u$.	$0, 2 \rightarrow 4$; Lemma 3.1.
High three-depth	$q = 3^i 5^j r$, $i \geq 2$, maximal among active orders divisible by nine.	Residual boundary or essential mass six exhausts all entries; mass five forces the sixth into order q .	Not used; maximality isolates the primitive q -row.	(4) , (5) , or $A = L_5(c) \uplus \{-5c\}$.	$2, 2 \rightarrow 6 \rightarrow 4$, standard cycle on X_m^4 , or $4, 2 \rightarrow 8 \rightarrow 4$; Lemmas 3.1–3.3, Theorem 7.1.
High five-depth, after high three	Maximal active base $h = 5^j r$, $j \geq 2$; faces $h, 3h$.	Residual boundary exhausts twelve. After $C = 0$, $k = 4$ and variation four exhaust twelve; $k = 2$ gives a four-entry core.	Omitted seven is used only after the residual boundary exhausts the ledger.	Standard-five; $A = B \uplus \delta_u$; or $S_c \uplus S_{3c} = A \uplus \delta_{3dc}$.	Standard cycle, $0, 2 \rightarrow 4$, or $2, 2 \rightarrow 6 \rightarrow 4$; Lemmas 3.1–3.3, Theorem 7.3.
Common content	All active orders now have three- and five-depth at most one.	No exhaustion needed; valuation bounds force common divisibility.	Not used.	$A = cA_0$, with c from Corollary 7.4; strict descent only for $c > 1$.	$X_m^4 \rightarrow X_{m/c}^4$; scalar c^5 in (2).
No small prime; or only D at the chosen large part	Maximal active exact order q , or maximal active large part $r > 1$.	The only possible nonzero odd kernel is a pair of opposite U_7 -fibres, of norm twelve.	Omitted seven is isolated after equality empties every other stratum.	Inverse-pair-free case impossible; without small primes, $A = \biguplus_{i=1}^3 \delta_{u_i}$.	$0, 2 \rightarrow 4$ for inverse pairs; Lemma 3.1, Proposition 10.1 and Section 10.

Continued on the next page.

Table 1 continued.

Branch	Maximal active order	Twelve-unit budget	When omitted-factor rows are available	Actual multiset identity or outcome	Source–target; nonvanishing
One small face, $e = 4$	Maximal active large part $r > 1$; faces sr, r .	Eight units form an actual Hodge quadruple; its two complements are inverse.	No omitted-large-factor row is needed for this ambient completion.	$A = B \uplus \delta_u$.	$0, 2 \rightarrow 4$; Lemma 3.1, Theorem 7.5.
One small face, $e = 5$	$s = 3, r = 11$, selected as a maximal active large part.	Ten units form the prefix; the zero-sum equation forces the sixth entry, of order three.	No omitted-large-factor row is used before actual completion.	$A = (m/33)w$, up to units and permutation; $a \uplus \delta_0 = Q \uplus S$, $S \uplus \delta_1 = \beta \uplus \gamma$.	At 66: $2, 2 \rightarrow 6 \rightarrow 4$, then $4, 2 \rightarrow 8 \rightarrow 4$; $X_{66}^4 \rightarrow X_{33}^4$ has scalar 32; Section 9.2, then (2).
One small face, $e = 6$	Maximal active large part $r > 1$; prime cases $r = p$.	$\ F_{sr}\ _1 + \ F_r\ _1 = 12$; all other strata empty.	Effective remaining rows are now available. The earlier paired-fibre alternative uses omitted seven only after $\ F_{sr}\ _1 = 12$.	Standard-five; $S_c \uplus S_{3c} = A \uplus \delta_{3dc}$; or the Q_p , split-triple, and level-33 identities of Section 9.2.	Standard cycle; $2, 2 \rightarrow 6 \rightarrow 4$; divisor on $X_m^1 \times X_m^1 \rightarrow X_m^4$ via Ran [6, Proposition 4.6, p. 140]; or the 66-to-33 certificate.
Mixed $F_{35} \neq 0$	Maximal active large part $r > 1$; faces $15r, 3r, 5r, r$.	Exceptional top kernels cost twelve. Nonconstant C forces equality in (21). Surviving column norm four with variation four exhausts twelve.	Lemma 8.1: omitted seven after top exhaustion. Lemma 8.3: residual row after equality in (21). Earlier actual completions need neither.	(5); $A = B \uplus \delta_u$; or the two-surface contraction identity of Theorem 7.3.	$4, 2 \rightarrow 8 \rightarrow 4$, $0, 2 \rightarrow 4$, or $2, 2 \rightarrow 6 \rightarrow 4$; Lemmas 3.1–3.3 and 8.1–8.4.
Mixed $F_{35} = 0$, both small faces nonzero	Maximal active large part $r > 1$.	The faces cost at least four and eight; equality makes each b_s one antipodal delta and empties all other strata.	All residual rows only after equality; exact-conductor- r rows and Corollary 2.3 give (22).	$r = 7$; the level-105 representative is $(L_5(3) \setminus \{45\}) \uplus \{50, 85\}$, as displayed in Section 10.	$4, 2 \rightarrow 8 \rightarrow 4$ via (5); Lemmas 3.1–3.3; [3, Theorem 1.3, p. 4].
Maximal active $r = 1$	Every active order divides s , or divides 15 when both small primes occur; ambient M need not be one.	No further exhaustion; common content removes the inactive ambient large part.	Not used.	Prime levels 3 and 5 impossible; at 15, standard-five or $(1, 2, 7, 11, 12, 12) \uplus (6, 9) = C(2) \uplus C(6)$.	Standard cycle or $2, 2 \rightarrow 6 \rightarrow 4$ at 15; Lemmas 3.1–3.3, Section 9, and transport by (2).

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